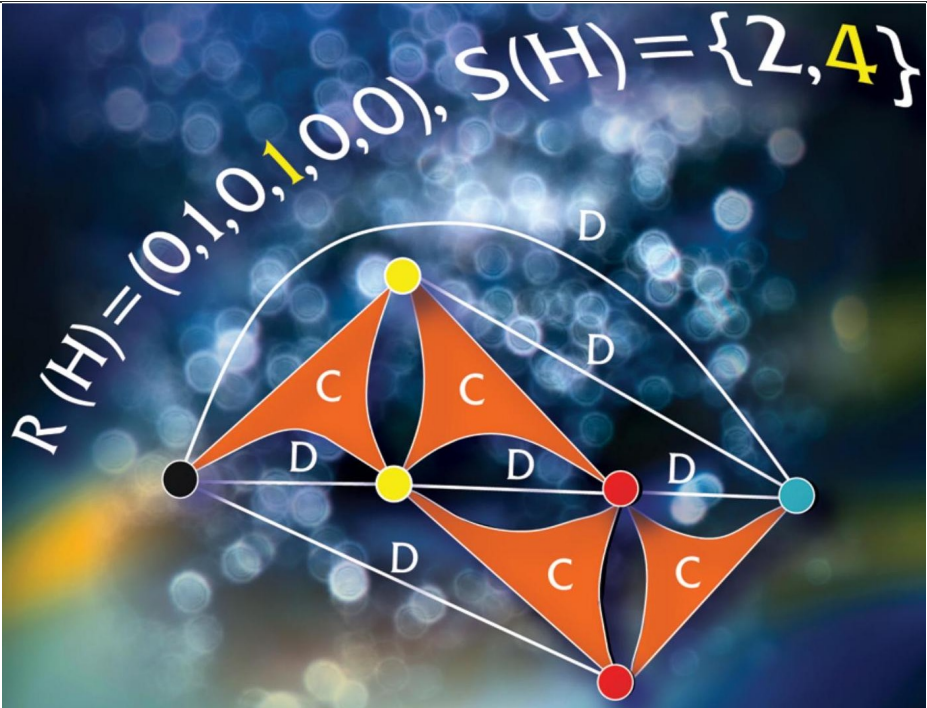
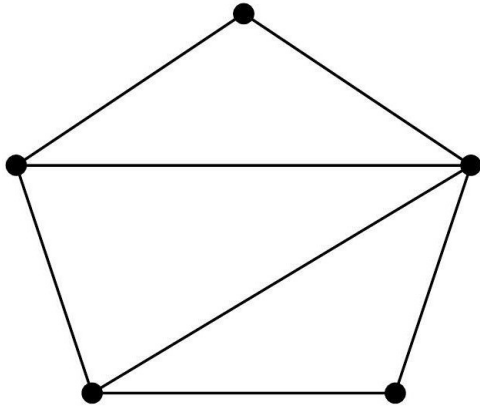
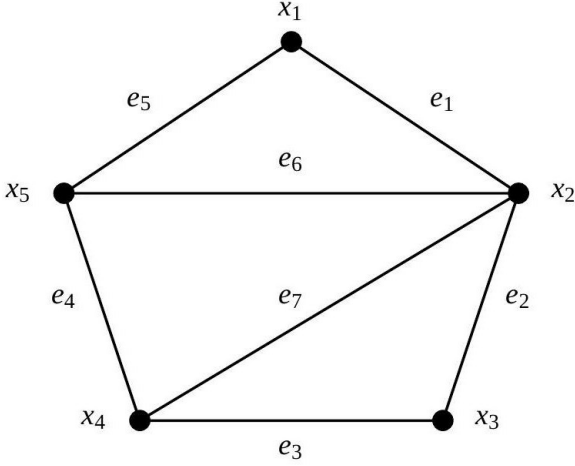
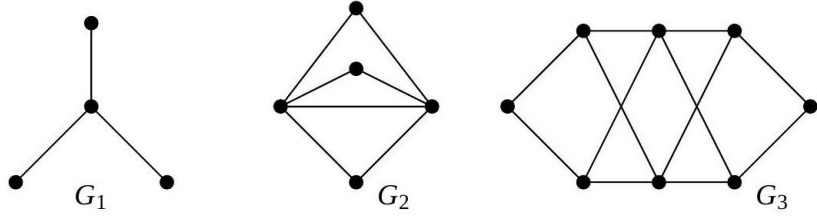
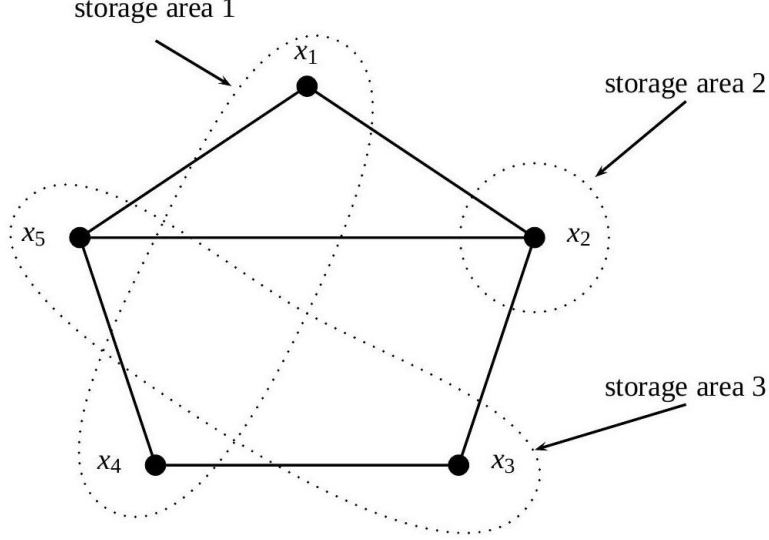
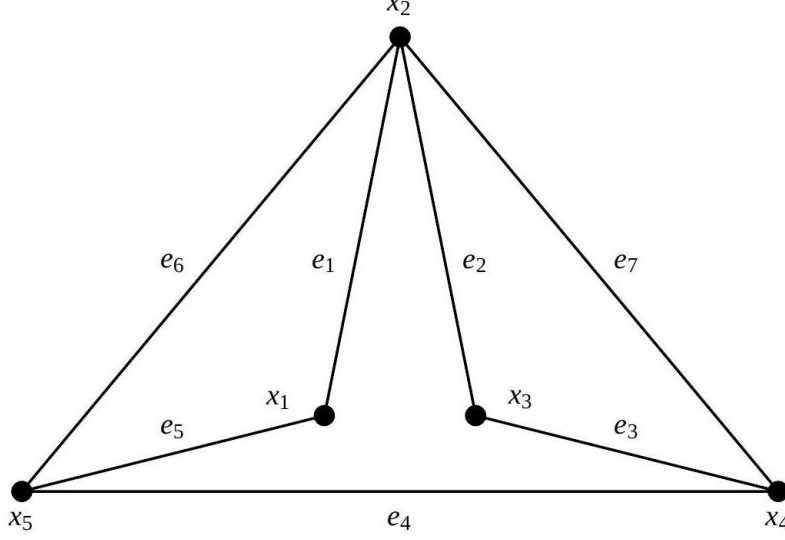
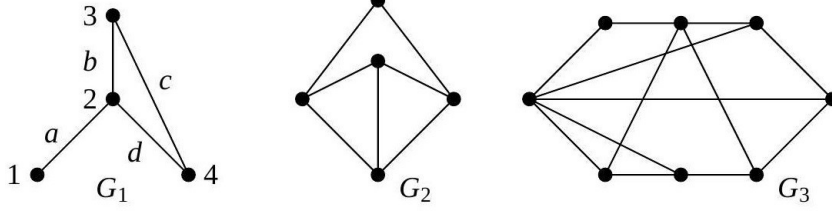
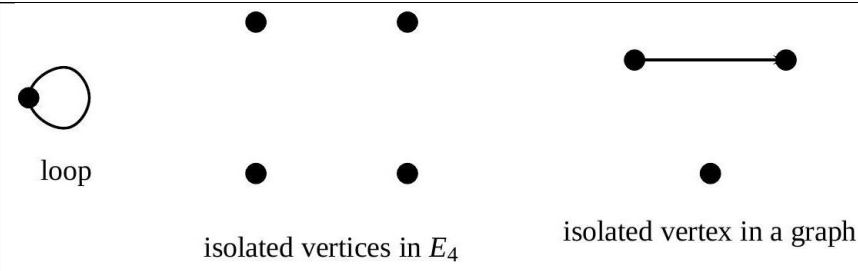
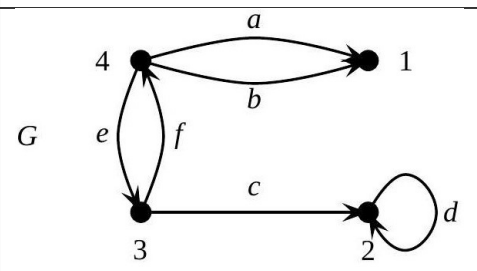
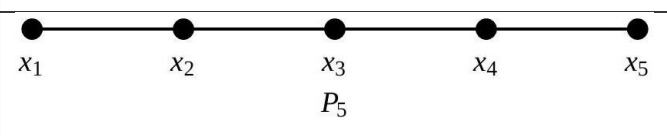
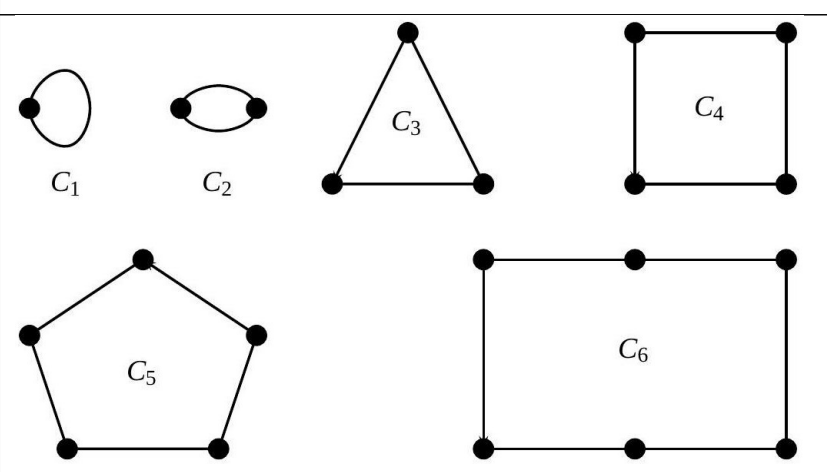
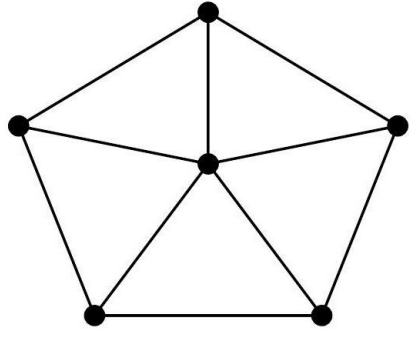
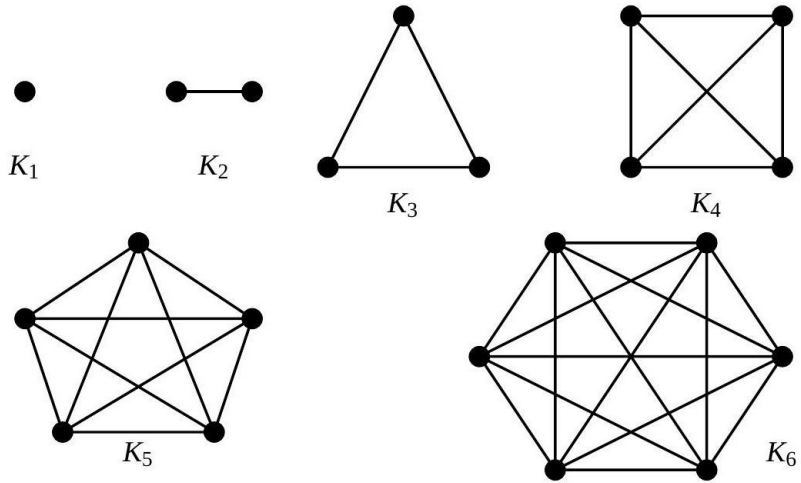
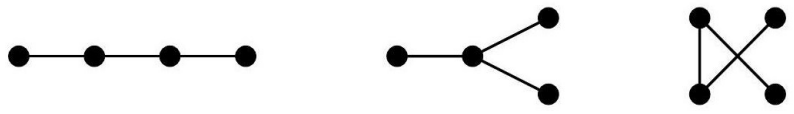
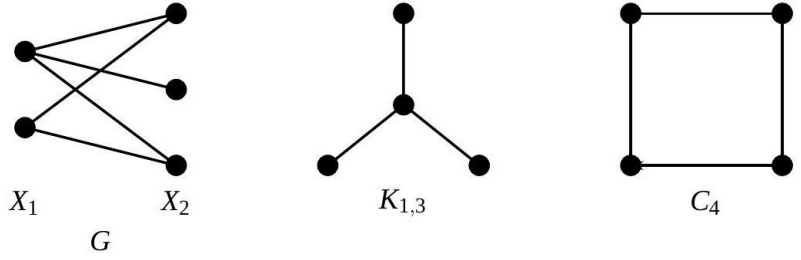
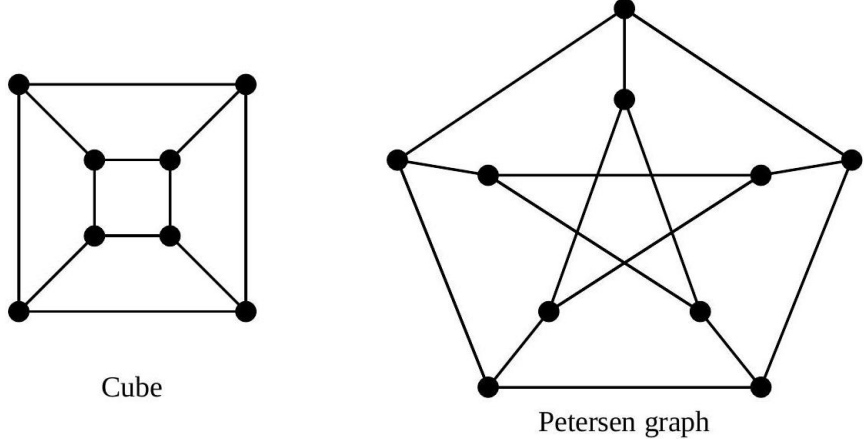


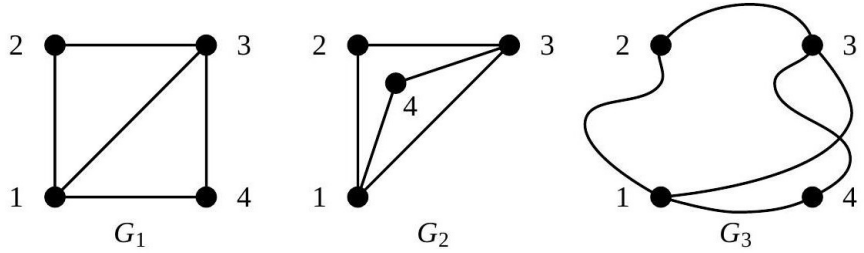
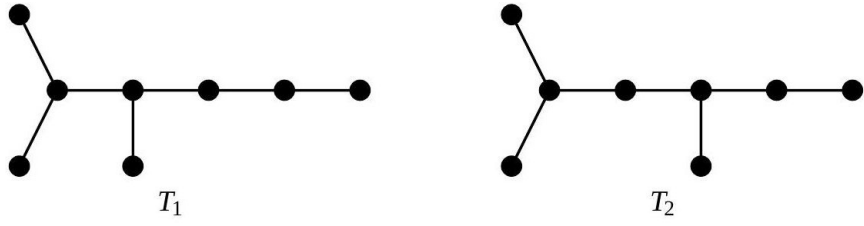
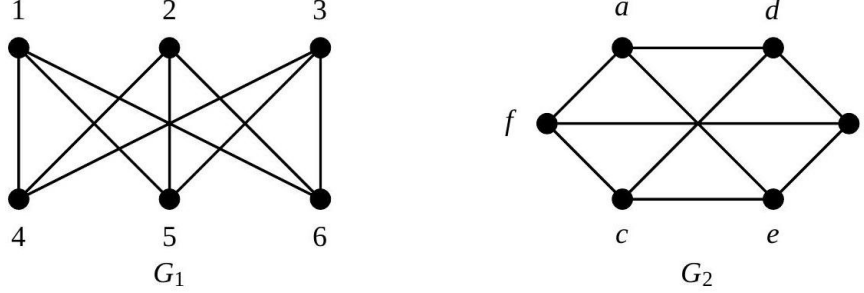
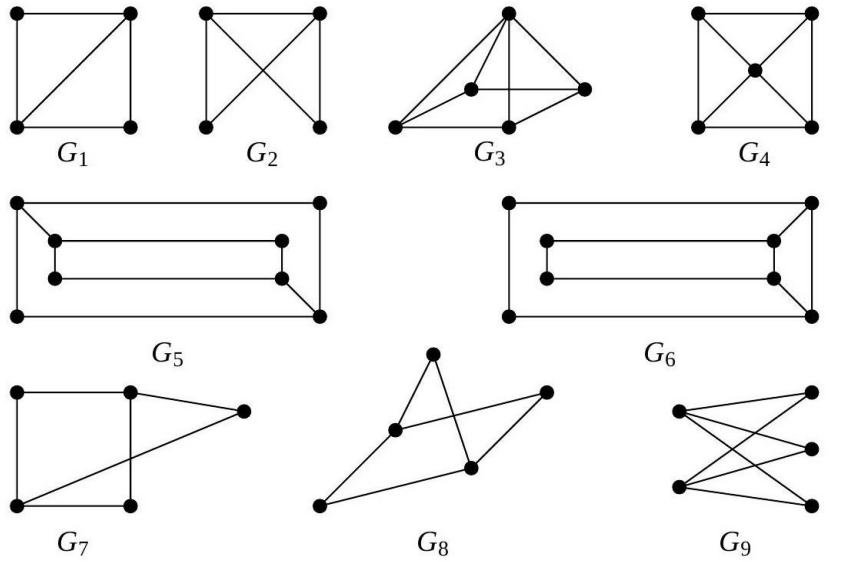
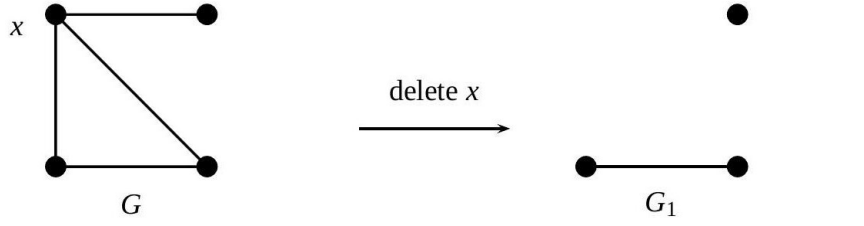
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voloshin-hypergraph_ DIA_0003	22	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0006	29	—	—	—	
voloshin-hypergraph_ DIA_0007	31	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0008	32	—	—	—	 loop isolated vertices in E_4 isolated vertex in a graph multiple edges arcs
voloshin-hypergraph_ DIA_0009	32	—	—	—	
voloshin-hypergraph_ DIA_0010	33	—	—	—	 G
voloshin-hypergraph_ DIA_0011	34	—	—	—	 P_5
voloshin-hypergraph_ DIA_0012	34	—	—	—	 G_1 G_2
voloshin-hypergraph_ DIA_0013	35	—	—	—	 C_1 C_2 C_3 C_4 C_5 C_6

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0017	37	—	—	—	
voloshin-hypergraph_ DIA_0018	38	—	—	—	

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voloshin-hypergraph_ DIA_0021	40	—	—	—	
voloshin-hypergraph_ DIA_0022	41	—	—	—	
voloshin-hypergraph_ DIA_0023	42	—	—	—	

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voloshin-hypergraph_ DIA_0026	43	—	—	—	
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voloshin-hypergraph_ DIA_0028	44	—	—	—	

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voloshin-hypergraph_ DIA_0031	48	—	—	—	
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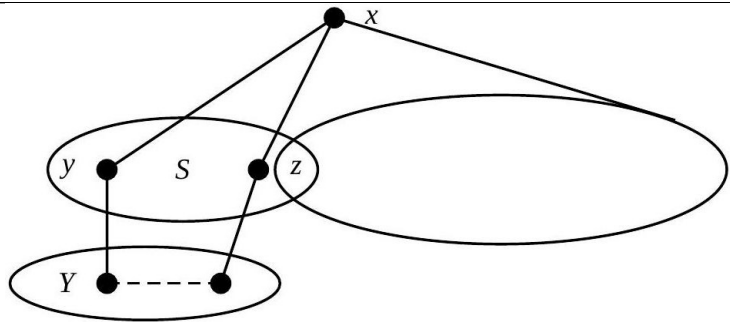
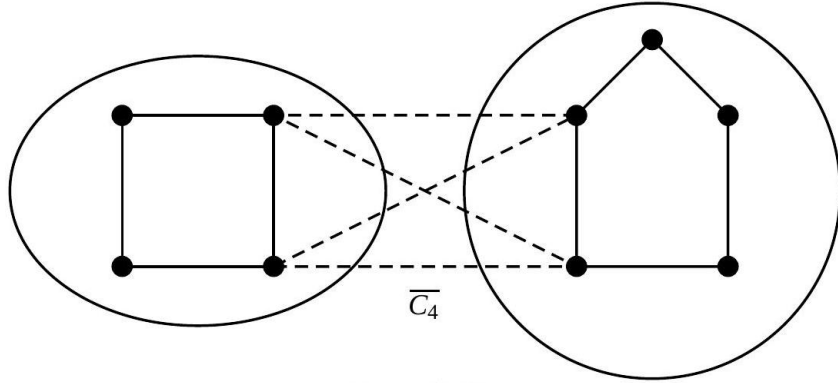
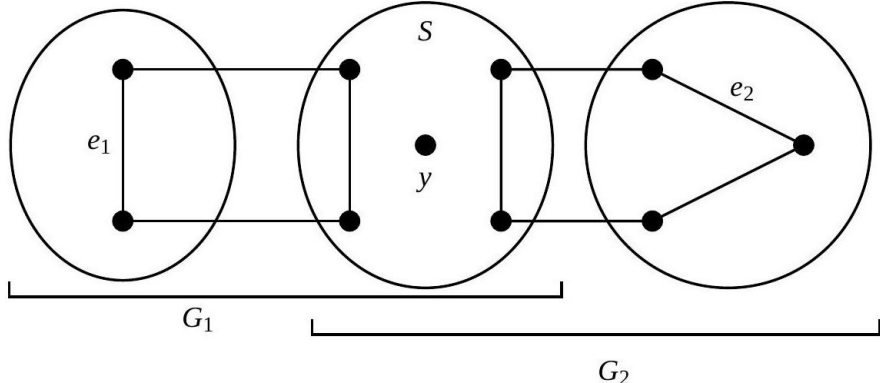
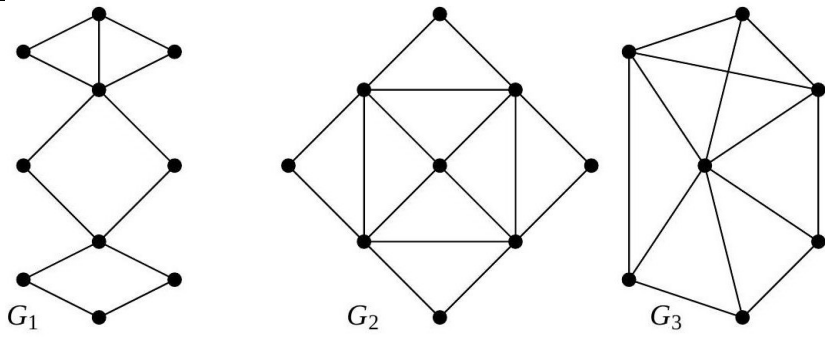
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voloshin-hypergraph_ DIA_0037	53	—	—	—	
voloshin-hypergraph_ DIA_0038	56	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0040	59	—	—	—	
voloshin-hypergraph_ DIA_0041	60	—	—	—	
voloshin-hypergraph_ DIA_0042	61	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0043	64	—	—	—	The diagram illustrates a relationship between two sets, X and Y. Set X is on the left and contains two subsets: S and $X - S_0$. Set Y is on the right and contains two subsets: $N_G(S)$ and $N_G(S_0)$. Two edges, G_A and G_B, are shown. Edge G_A connects the subset S in X to the subset $N_G(S_0)$ in Y. Edge G_B connects the subset $X - S_0$ in X to the subset $N_G(S)$ in Y.
voloshin-hypergraph_ DIA_0044	65	—	—	—	The diagram shows two sets, X and Y. Set X contains subsets S and A. Set Y contains subsets $N_{G_1}(S)$ and B. A horizontal dashed line divides the diagram into two regions: G_1 above and G_2 below. Two edges are shown: one connecting S to $N_{G_1}(S)$ and another connecting A to B.
voloshin-hypergraph_ DIA_0045	66	—	—	—	The diagram depicts a grid of points connected by lines. There are four rows and six columns of points. Horizontal lines connect adjacent points in each row. Vertical lines connect adjacent points in each column. This structure forms a rectangular grid with internal connections.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0046	68	—	—	—	
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voloshin-hypergraph_ DIA_0048	70	—	—	—	
voloshin-hypergraph_ DIA_0049	72	—	—	—	
voloshin-hypergraph_ DIA_0050	74	—	—	—	
voloshin-hypergraph_ DIA_0051	75	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0054	76	—	—	—	
voloshin-hypergraph_ DIA_0055	77	—	—	—	
voloshin-hypergraph_ DIA_0056	78	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0059	81	—	—	—	
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
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voloshin-hypergraph_ DIA_0062	84	—	—	—	
voloshin-hypergraph_ DIA_0063	86	—	—	—	
voloshin-hypergraph_ DIA_0064	87	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0065	88	—	—	—	The diagram illustrates the construction of a hypergraph from two base graphs, K_5 and $K_{3,3}$. Top section (K_5): A complete graph K_5 with vertices 1, 2, 3, 4, 5 is shown. It is transformed into a triangle with vertices 1, 2, 3. Vertex 4 is placed either 'inside' or 'outside' the triangle. In both cases, the resulting graph has 'no face for 5'. Bottom section ($K_{3,3}$): A complete bipartite graph $K_{3,3}$ with vertices 1, 2, 3, 4, 5, 6 is shown. It is transformed into a square with vertices 1, 2, 3, 4. Vertex 5 is placed either 'inside' or 'outside' the square. In both cases, the resulting graph has 'no face for 6'.
voloshin-hypergraph_ DIA_0066	91	—	—	—	Two graphs, G_1 and G_2, are shown. G_1 is a complex graph with 10 vertices and many edges. G_2 is a graph with 10 vertices and 12 edges, including a cycle of length 6.
voloshin-hypergraph_ DIA_0067	92	—	—	—	A graph with 10 vertices and 12 edges, including a cycle of length 6.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0068	93	—	—	—	
voloshin-hypergraph_ DIA_0069	97	—	—	—	
voloshin-hypergraph_ DIA_0070	99	—	—	—	
voloshin-hypergraph_ DIA_0071	100	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0072	101	—	—	—	Diagram illustrating the decomposition of a hypergraph G into two hypergraphs G_1 and G_2. G has two vertices x and y and a hyperedge containing both. G_1 has a dashed edge between x and y. G_2 has a single vertex xy.
voloshin-hypergraph_ DIA_0073	102	—	—	—	Diagram illustrating the decomposition of a graph G into two graphs G_1 and G_2. G is a V-shaped graph with vertices x, y, and z. $G_1 = K_3$ is a triangle with vertices x, y, and z. $G_2 = K_2$ is a single edge with vertices x and z.
voloshin-hypergraph_ DIA_0074	103	—	—	—	Diagram illustrating a tree structure G with root G and children G_1 and G_2. G_1 has children G_3 and G_4. G_2 has children G_5 and G_6. Below the tree is a sequence of labels K_{i_1}, K_{i_2}, ..., K_{i_s}.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0075	104	—	—	—	
voloshin-hypergraph_ DIA_0076	105	—	—	—	
voloshin-hypergraph_ DIA_0077	107	—	—	—	

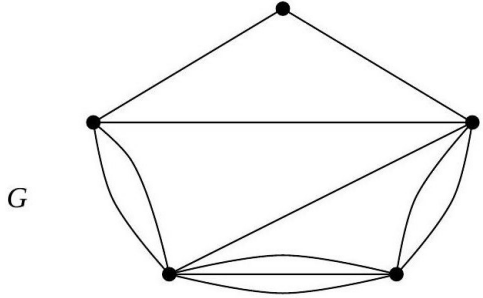
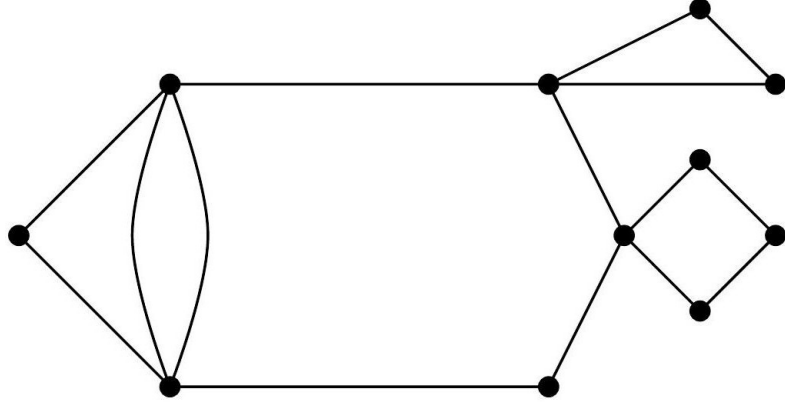
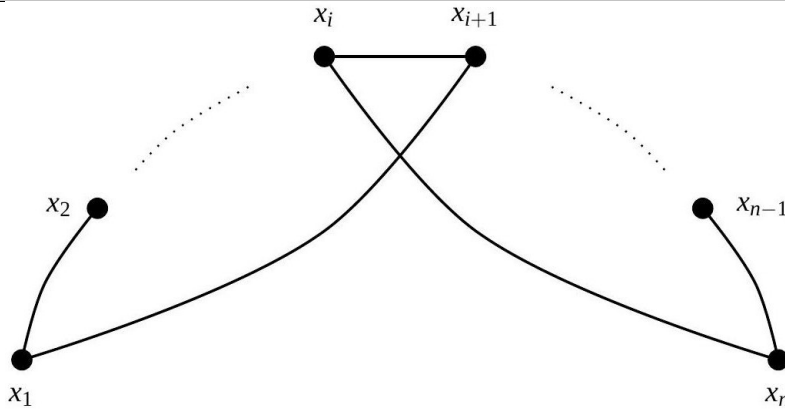
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0078	108	—	—	—	
voloshin-hypergraph_ DIA_0079	109	—	—	—	
voloshin-hypergraph_ DIA_0080	111	—	—	—	
voloshin-hypergraph_ DIA_0081	112	—	—	—	
voloshin-hypergraph_ DIA_0082	113	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0083	115	—	—	—	
voloshin-hypergraph_ DIA_0084	117	—	—	—	
voloshin-hypergraph_ DIA_0085	119	—	—	—	
voloshin-hypergraph_ DIA_0086	120	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0087	121	—	—	—	
voloshin-hypergraph_ DIA_0088	122	—	—	—	
voloshin-hypergraph_ DIA_0089	124	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0090	125	—	—	—	The diagram shows a transformation from a single edge (two vertices connected by a line) to a more complex graph structure. The transformation is indicated by an arrow pointing to the right. The resulting graph has five vertices and several edges, with one edge highlighted by a dashed circle.
voloshin-hypergraph_ DIA_0091	125	—	—	—	The diagram shows a complex graph structure with many vertices and edges, forming a dense network. The vertices are arranged in a roughly circular pattern, with many edges connecting them.
voloshin-hypergraph_ DIA_0092	126	—	—	—	The diagram shows two graph structures, G and $\bar{G}$, with vertices labeled 1, 2, 3. The vertices are arranged in a roughly circular pattern, with edges connecting them. The graph G is on the left, and $\bar{G}$ is on the right.
voloshin-hypergraph_ DIA_0093	129	—	—	—	The diagram shows two graph structures, G and $L(G)$, with vertices labeled a, b, c, d, e. The vertices are arranged in a roughly circular pattern, with edges connecting them. The graph G is on the left, and $L(G)$ is on the right. edge coloring vertex coloring

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0094	130	—	—	—	
voloshin-hypergraph_ DIA_0095	131	—	—	—	
voloshin-hypergraph_ DIA_0096	133	—	—	—	
voloshin-hypergraph_ DIA_0097	134	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0098	136	—	—	—	
voloshin-hypergraph_ DIA_0099	140	—	—	—	
voloshin-hypergraph_ DIA_0100	141	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0101	145	—	—	—	The image displays three network flow diagrams arranged vertically, connected by downward arrows, illustrating the progression of a flow augmentation algorithm. Each diagram features a source node u and a sink node v, with intermediate nodes x_1, x_2, x_3, x_4. Initial flow F_0: - Edges and flows: $u \rightarrow x_1$ (0,3), $u \rightarrow x_4$ (1,5), $x_1 \rightarrow x_2$ (1,7), $x_1 \rightarrow x_3$ (1,1), $x_2 \rightarrow v$ (1,6), $x_3 \rightarrow v$ (0,2), $x_4 \rightarrow x_3$ (1,4). Flow F_1: - Edges and flows: $u \rightarrow x_1$ (1,3), $u \rightarrow x_4$ (1,5), $x_1 \rightarrow x_2$ (1,7), $x_1 \rightarrow x_3$ (0,1), $x_2 \rightarrow v$ (1,6), $x_3 \rightarrow v$ (1,2), $x_4 \rightarrow x_3$ (1,4). Flow F_2: - Edges and flows: $u \rightarrow x_1$ (3,3), $u \rightarrow x_4$ (1,5), $x_1 \rightarrow x_2$ (3,7), $x_1 \rightarrow x_3$ (0,1), $x_2 \rightarrow v$ (3,6), $x_3 \rightarrow v$ (1,2), $x_4 \rightarrow x_3$ (1,4).

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0102	146	—	—	—	
voloshin-hypergraph_ DIA_0103	151	—	—	—	
voloshin-hypergraph_ DIA_0104	155	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image																																										
voloshin-hypergraph_ DIA_0105	157	—	—	—	The diagram illustrates the relationship between a hypergraph $\mathcal{H} = (X, \mathcal{D})$, its incidence matrix $I(\mathcal{H})$, and its bipartite graph $B(\mathcal{H})$. Hypergraph $\mathcal{H}$: A set of 6 vertices $\{1, 2, 3, 4, 5, 6\}$ and 5 hyperedges D_1, D_2, D_3, D_4, D_5. Vertex 1 is in D_1. Vertices 1, 2, 3, 4, 5 are in D_2. Vertices 1, 2, 3, 5, 6 are in D_3. Vertices 1, 2, 3, 4 are in D_4. Vertices 1, 3, 4, 5, 6 are in D_5. Incidence Matrix $I(\mathcal{H})$: <table border="1"> <thead> <tr> <th></th><th>D_1</th><th>D_2</th><th>D_3</th><th>D_4</th><th>D_5</th></tr> </thead> <tbody> <tr> <th>1</th><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr> <tr> <th>2</th><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr> <tr> <th>3</th><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr> <tr> <th>4</th><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr> <tr> <th>5</th><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr> <tr> <th>6</th><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> </tbody> </table> Bipartite Graph $B(\mathcal{H})$: A bipartite graph with vertices $\{1, 2, 3, 4, 5, 6\}$ on the left and $\{D_1, D_2, D_3, D_4, D_5\}$ on the right. Edges connect vertices to the hyperedges they belong to: $(1, D_1), (1, D_2), (1, D_3), (2, D_2), (2, D_3), (2, D_4), (3, D_2), (3, D_3), (3, D_4), (3, D_5), (4, D_2), (4, D_3), (4, D_4), (4, D_5), (5, D_2), (5, D_3), (5, D_4), (5, D_5)$. Hypergraph $\mathcal{H}^*$: A set of 6 vertices $\{1, 2, 3, 4, 5, 6\}$ and 5 hyperedges d_1, d_2, d_3, d_4, d_5. Vertex 1 is in d_1. Vertices 1, 2, 3, 4, 5 are in d_2. Vertices 1, 2, 3, 5, 6 are in d_3. Vertices 1, 2, 3, 4 are in d_4. Vertices 1, 3, 4, 5, 6 are in d_5.		D_1	D_2	D_3	D_4	D_5	1	1	1	1	0	0	2	0	1	1	1	0	3	0	0	0	1	1	4	0	0	1	0	1	5	0	0	0	1	1	6	0	0	0	0	0
	D_1	D_2	D_3	D_4	D_5																																										
1	1	1	1	0	0																																										
2	0	1	1	1	0																																										
3	0	0	0	1	1																																										
4	0	0	1	0	1																																										
5	0	0	0	1	1																																										
6	0	0	0	0	0																																										
voloshin-hypergraph_ DIA_0106	158	—	—	—	The diagram illustrates the relationship between a graph G and its dual graph G^*. Graph G: A graph with 5 vertices $\{1, 2, 3, 4, 5\}$ and 5 edges a, b, c, d, e. Edges a, b, c, d, e form a cycle $1-2-3-4-5-1$. Vertex 3 is connected to vertex 2 by edge c. Dual Graph G^*: A graph with 5 vertices $\{a, b, c, d, e\}$ and 5 edges $1, 2, 3, 4, 5$. Edges $1, 2, 3, 4, 5$ form a cycle $a-b-c-d-e-a$. Vertex c is connected to vertex b by edge 3.																																										

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0107	161	—	—	—	Diagrams illustrating various hypergraph types K_4^i for $i = 0, 1, 2, 3, 4$. K_4^0 is a single vertex. K_4^1 is a single edge (two vertices). K_4^2 is a square with both diagonals. K_4^3 is a complete graph K_4 with four vertices and six edges. K_4^4 is a single hyperedge containing all four vertices.
voloshin-hypergraph_ DIA_0108	163	—	—	—	Diagram illustrating the strong deletion of a vertex x from a hypergraph $\mathcal{H}$. The left side shows $\mathcal{H}$ with vertex x (circled) and its incident edges. An arrow labeled "delete x strongly" points to the resulting hypergraph $\mathcal{H}_1 = \mathcal{H} - x$, where x and its incident edges are removed.
voloshin-hypergraph_ DIA_0109	163	—	—	—	Diagram illustrating the weak deletion of a vertex x from a hypergraph $\mathcal{H}$. The left side shows $\mathcal{H}$ with vertex x (circled) and its incident edges. An arrow labeled "delete x weakly" points to the resulting hypergraph $\mathcal{H}_1 = \mathcal{H} - x$, where x is removed but its incident edges remain as separate hyperedges.
voloshin-hypergraph_ DIA_0110	164	—	—	—	Diagram illustrating the strong deletion of a set D from a hypergraph $\mathcal{H}$. The left side shows $\mathcal{H}$ with set D (circled) and its incident edges. An arrow labeled "delete D strongly" points to the resulting hypergraph $\mathcal{H}_1 = \mathcal{H} - D$, where D and its incident edges are removed.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0111	164	—	—	—	
voloshin-hypergraph_ DIA_0112	165	—	—	—	
voloshin-hypergraph_ DIA_0113	166	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0114	168	—	—	—	
voloshin-hypergraph_ DIA_0115	171	—	—	—	
voloshin-hypergraph_ DIA_0116	172	—	—	—	

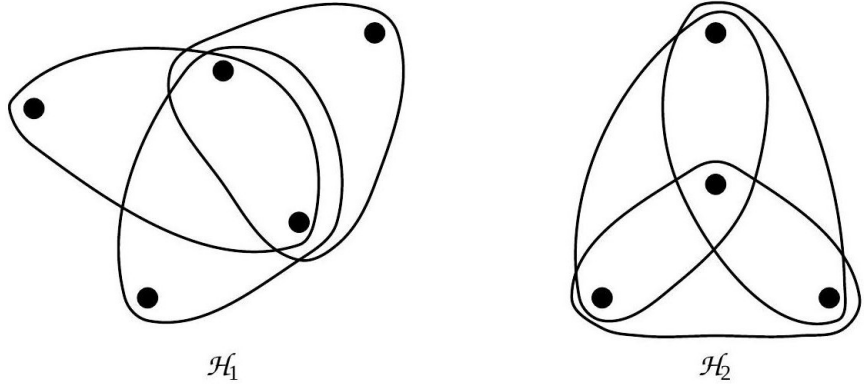
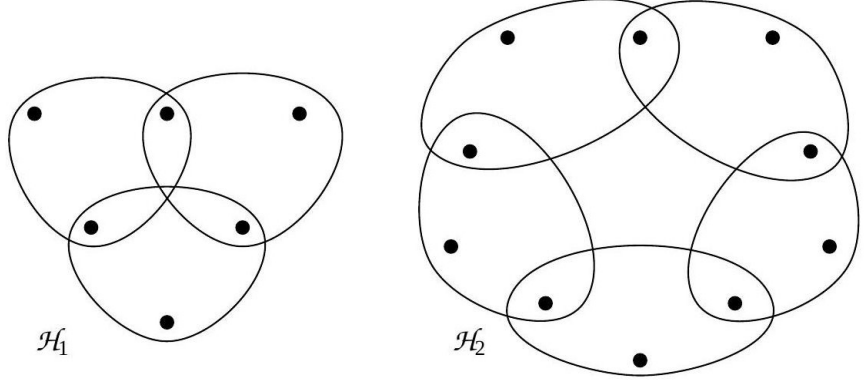
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0117	174	—	—	—	$\mathcal{H} = K_3$ not conformal, not Helly $\mathcal{H}_1$ - conformal, not Helly $\mathcal{H}_2 \cong \mathcal{H}_1^*$ - Helly, not conformal $\mathcal{H}_3 \cong \mathcal{H}_3^*$ - Helly, conformal
voloshin-hypergraph_ DIA_0118	178	—	—	—	T C_4

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0119	179	—	—	—	$\mathcal{H}_1$ $\mathcal{H}_2$
voloshin-hypergraph_ DIA_0120	180	—	—	—	$\mathcal{H}$ $\mathcal{H}_0$
voloshin-hypergraph_ DIA_0121	181	—	—	—	$\mathcal{H}_0^*$ $\mathcal{H}^*$
voloshin-hypergraph_ DIA_0122	181	—	—	—	$\mathcal{H}_0^*$ $\mathcal{H}^*$
voloshin-hypergraph_ DIA_0123	182	—	—	—	G $\mathcal{H}$ $\mathcal{H}^*$

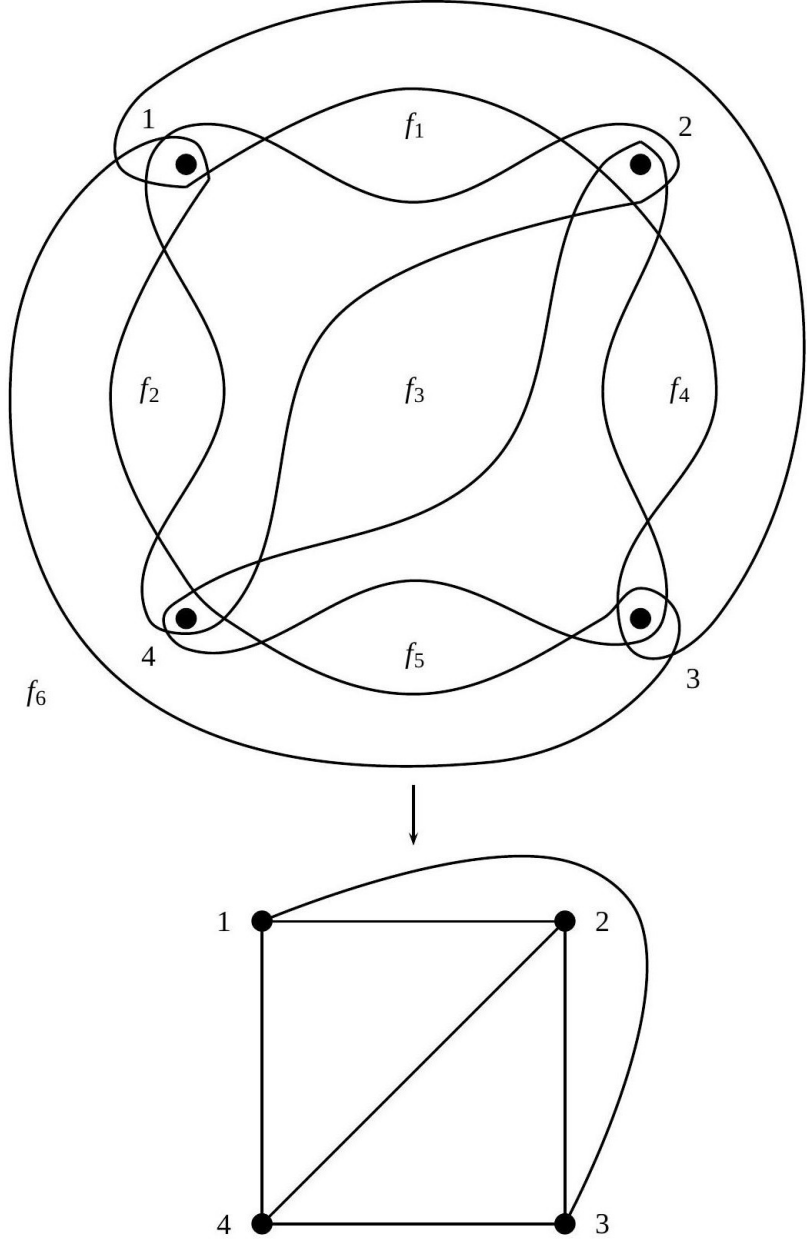
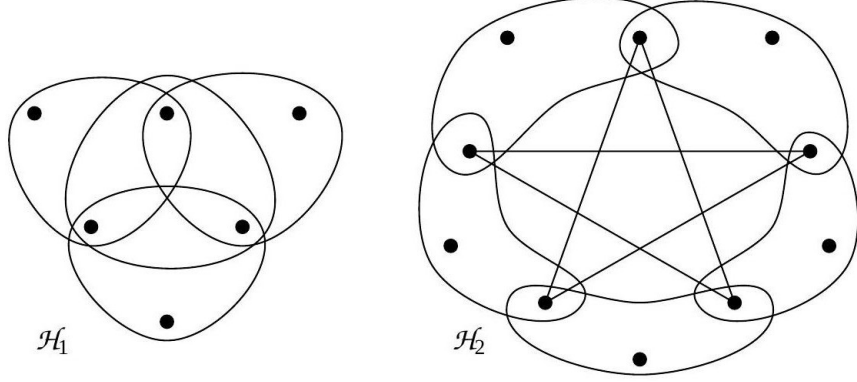
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0124	183	—	—	—	
voloshin-hypergraph_ DIA_0125	186	—	—	—	
voloshin-hypergraph_ DIA_0126	187	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0127	188	—	—	—	
voloshin-hypergraph_ DIA_0128	189	—	—	—	
voloshin-hypergraph_ DIA_0129	191	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0130	194	—	—	—	Diagram illustrating a hypergraph $\mathcal{H}$ and its associated tree $T \subseteq [\mathcal{H}]_2$. The top part shows $\mathcal{H}$ with a vertex x and an arrow pointing to T. The bottom part shows $\mathcal{H} - x$ and an arrow pointing to $T - x$.
voloshin-hypergraph_ DIA_0131	196	—	—	—	Diagram illustrating two hypergraphs $\mathcal{H}_1$ and $\mathcal{H}_2$.
voloshin-hypergraph_ DIA_0132	198	—	—	—	Diagram illustrating a hypergraph structure with eight vertices and four hyperedges.
voloshin-hypergraph_ DIA_0133	199	—	—	—	Diagram illustrating two hypergraph structures.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0134	200	—	—	—	 $\mathcal{H}_1$ $\mathcal{H}_2$
voloshin-hypergraph_ DIA_0135	202	—	—	—	 $\mathcal{H}_1$ $\mathcal{H}_2$

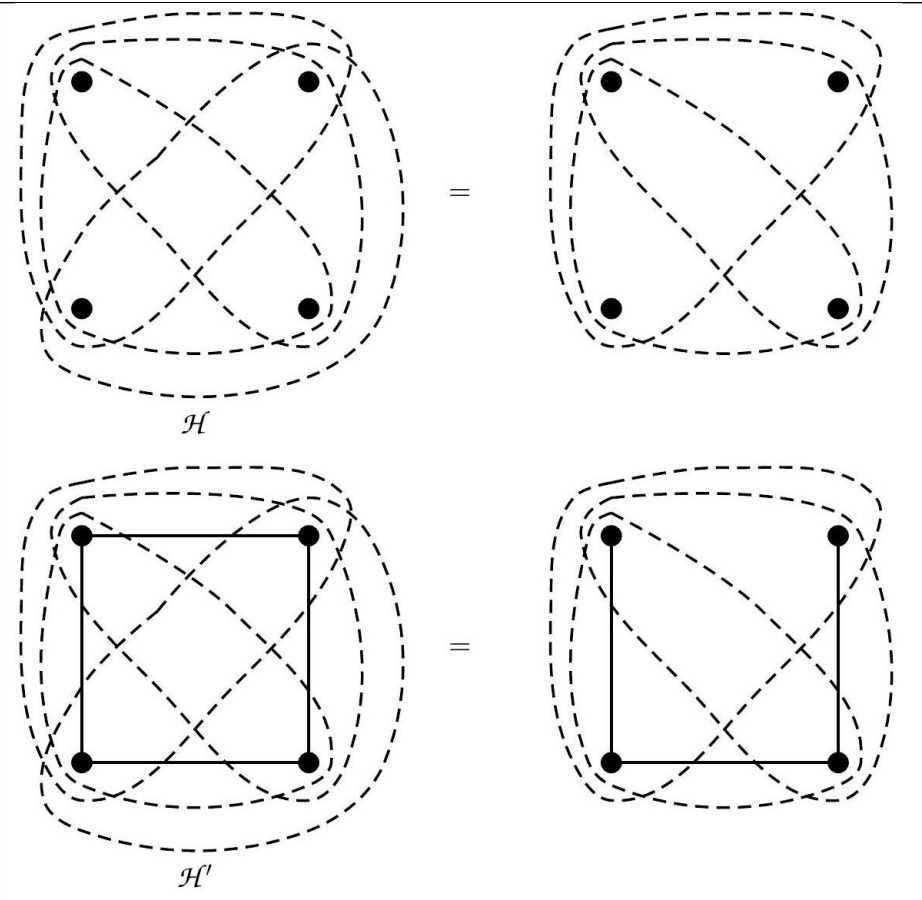
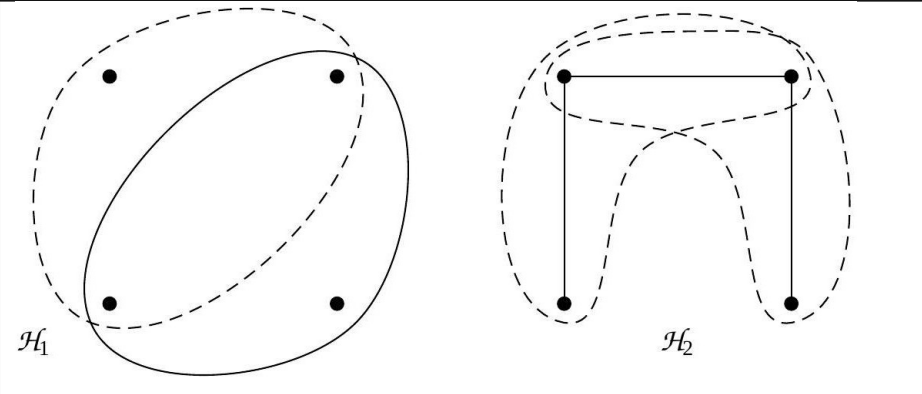
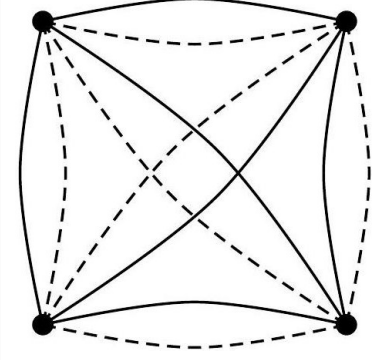
Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0136	203	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0137	206	—	—	—	
voloshin-hypergraph_ DIA_0138	207	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0139	210	—	—	—	
voloshin-hypergraph_ DIA_0140	213	—	—	—	
voloshin-hypergraph_ DIA_0141	216	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0147	226	—	—	—	
voloshin-hypergraph_ DIA_0148	229	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0149	232	—	—	—	The diagram illustrates a sequence of operations on a hypergraph C with nodes $1, 2, 3, 4$ and hyperedges D_1, D_2, D_3, D_4. Top Row: C is shown as a diamond shape with nodes $1, 2, 3, 4$ and hyperedges D_1, D_2, D_3, D_4. This is equated to the sum of two hypergraphs: one with a vertical hyperedge 13 and one with a vertical dashed hyperedge 13. Second Row: The first hypergraph from the top row is split (spl. 24) into two hypergraphs: one with a horizontal hyperedge 24 and one with a horizontal dashed hyperedge 24. The second hypergraph from the top row is processed by C-clearing + contraction 13 to form a triangle with nodes $13, 4, 2$, which is then split (spl. 24) into two triangles: one with a horizontal hyperedge 24 and one with a horizontal dashed hyperedge 24. Third Row: The first hypergraph from the second row is processed by C-clearing + contraction 24 to form a triangle with nodes $13, 4, 2$. The second hypergraph from the second row is processed by C-clearing + contraction 24 to form a triangle with nodes $13, 4, 2$. The third hypergraph from the second row is processed by contraction 24 to form a triangle with nodes $13, 4, 2$. Bottom Row: The first hypergraph from the third row is processed by contraction 24 to form a triangle with nodes $13, 4, 2$. The second hypergraph from the third row is processed by contraction 24 to form a triangle with nodes $13, 4, 2$. The third hypergraph from the third row is processed by contraction 24 to form a triangle with nodes $13, 4, 2$.

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voloshin-hypergraph_ DIA_0150	234	—	—	—	
voloshin-hypergraph_ DIA_0151	235	—	—	—	
voloshin-hypergraph_ DIA_0152	237	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0153	238	—	—	—	
voloshin-hypergraph_ DIA_0154	239	—	—	—	
voloshin-hypergraph_ DIA_0155	242	—	—	—	
voloshin-hypergraph_ DIA_0156	242	—	—	—	$\mathcal{H}$

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voloshin-hypergraph_ DIA_0157	244	—	—	—	
voloshin-hypergraph_ DIA_0158	250	—	—	—	
voloshin-hypergraph_ DIA_0159	251	—	—	—	
voloshin-hypergraph_ DIA_0160	253	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0161	257	—	—	—	
voloshin-hypergraph_ DIA_0162	258	—	—	—	
voloshin-hypergraph_ DIA_0163	259	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0164	261	—	—	—	
voloshin-hypergraph_ DIA_0165	262	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0166	265	—	—	—	
voloshin-hypergraph_ DIA_0167	267	—	—	—	
voloshin-hypergraph_ DIA_0168	269	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0169	271	—	—	—	
voloshin-hypergraph_ DIA_0170	275	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0171	276	—	—	—	
voloshin-hypergraph_ DIA_0172	277	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0173	280	—	—	—	The diagram illustrates a mapping G from a simple graph to a hypergraph. The top graph has nodes Z_1, Z_2, Z_3 and edges $\{a, c\}, \{a, b\}, \{b, c\}$. The bottom hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ has nodes a, b, c and hyperedges Z_1, Z_2, Z_3. A double-headed arrow labeled G connects the two graphs.
voloshin-hypergraph_ DIA_0174	281	—	—	—	The diagram illustrates a mapping G from a bipartite graph to a hypergraph. The top graph has nodes x_1, x_2, x_3, x_4 (jobs) and y_1, y_2, y_3, y_4, y_5 (resources) with various edges. The bottom hypergraph $\mathcal{H} = (X, \mathcal{C}, \mathcal{D})$ has nodes x_1, x_2, x_3, x_4 and hyperedges Z_1, Z_2, Z_3. A double-headed arrow labeled G connects the two graphs.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
voloshin-hypergraph_ DIA_0175	284	—	—	—	Induction basis Induction hypothesis Induction step 1. P_1 2. $P_k \rightarrow P_{k+1}$ $\longrightarrow P_n$ is true for all $n = 1, 2, \dots$
voloshin-hypergraph_ DIA_0176	284	—	—	—	Induction basis Induction hypothesis Induction step 1. P_1 2. $P_1, P_2, \dots, P_k \rightarrow P_{k+1}$ $\longrightarrow P_n$ is true for all $n = 1, 2, \dots$